

STANDARD OPERATING PROCEDURE

SOP-ELEC-042: Power Transformer Maintenance

1. TEMPERATURE LIMITS

- Normal Top Oil Temperature: $< 75^{\circ}\text{C}$.
- Alert Top Oil Temperature: 75°C to 85°C . Action: Increase monitoring frequency and verify cooling fan operation.
- Trip Top Oil Temperature: $> 85^{\circ}\text{C}$. Action: Automatic trip / immediate shutdown.

2. DISSOLVED GAS ANALYSIS (DGA)

- Hydrogen (H_2) < 50 ppm: Normal operation.
- Hydrogen (H_2) > 100 ppm: Indicates potential partial discharge or corona. Action: Schedule an outage within 7 days for electrical testing.
- Acetylene (C_2H_2) > 5 ppm: Indicates active arcing. Action: Immediate emergency shutdown required.

3. SILICA GEL BREATHER

- Orange color indicates active moisture absorption (Good).
- Green or transparent color indicates moisture saturation.
- Action: Replace silica gel immediately when it turns green to prevent moisture ingress into the transformer oil.

4. OIL LEAKS

- Any observed oil seepage must be logged.
- Seepage without active dripping can be repaired during the next scheduled maintenance window.